Circuit investigation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements to test a circuit model safely.

SUCCESS CRITERIA - I can...

1. Draw and predict before powering a circuit.
2. Place ammeters in series and voltmeters in parallel.
3. Compare measured and predicted values using evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

An ideal series loop has supply 15 V. One component drops 3 V. Find the other component's drop.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: V = IR.
Use approved low voltage. Ammeter in series; voltmeter across a component. Disconnect before rewiring; never put an ammeter across the supply.

Worked reasoning: Supplied measurements show V=2, 4, 6 V and I=0.2, 0.4, 0.6 A. Each V/I is 10 ohms. This supports an ohmic model over the measured range.

Your turn: Plot V against I and mark the slope. State why the data do not prove the model at every possible temperature or voltage.

Marzano check - circle: ① ② ③

03 Look and explain

On a resistor circuit sketch, add an ammeter to measure loop current and a voltmeter to measure the resistor's drop.



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Build the solution

A supplied 6 V, 12 Ω ideal circuit predicts I=6÷12. Calculate it. Then prepare a table for actual voltage, actual current and one observation; leave measurements blank until the trial.

Voltage (V)	Current (A)	Observation

Marzano check - circle: ① ② ③

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Continue down each column.

05 Use the idea

A reading disagrees with your prediction. List two checks before replacing the model or deleting data.

Marzano check - circle: 1 2 3

06 Choose a method

An ideal blackbody’s kelvin temperature becomes 4 times as large. How does its peak wavelength change?

Marzano check - circle: 1 2 3

07 Exit check

Where must the teacher check your circuit before the first powered test? Record that checkpoint and one change suggested.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A later reading is 8 V and 0.5 A. Does it fit the same 10 ohm model? Name two checks before declaring the physical law wrong.

Marzano check - circle: 1 2 3